COURSE……………………………………………………………………………INDEX NO………………………………………………….…

KWAME NKRUMAH UNIVERSITY OF SCIENCE AND TECHNOLOGY, KUMASI

COLLEGE OF ENGINEERING

B.Sc. (Engineering), Mid-Semester Examination, March, 2020

First (1st) Year

Agric, Aerospace, Automobile, Biomedical, Computer, Chemical, Material and Mechanical Engineering

COE/EE 152: BASIC ELECTRONICS

TIME: 60 MINS

PROGRAM OF STUDY: ....

INDEX NO:………..........…………..

Instructions

i. Candidates are to indicate their Index Number and Program of Study in the spaces provided.
ii. Candidates are to submit <u>ONLY</u> the Scannable Sheet.
iii. Choose from the options lettered A to D, the most suitable answer to the question statements
iv. Shade the most suitable answer to the question statements on the Scannable Sheet provided
v. Candidates are to write the answer at the back of the Scannable Sheet if he/she thinks the right answer cannot be found in the options provided
vi. Each correct answer carries 0.5 a mark

<u>CANDIDATES ARE ALLOWED TO TAKE THE QUESTION PAPERS AWAY</u>

1. When used in a circuit, a zener diode is always
(a) at a temperature below 0 °C
(b) connected in series
(c) reverse-biased
(d) forward-biased
(e) None of the above.

2. Major part of electric current in an intrinsic semiconductor is due to drift of
(a) conduction –band electrons
(b) valence –band electrons
(c) conduction-band holes
(d) valence-band holes
(e) None of the above.

3. The rms value of current in a full-wave rectifier is given as
(a) I_m/π
(b) $I_m/2$
(c) $I_m/\sqrt{2}$
(d) $2I_m/\pi$

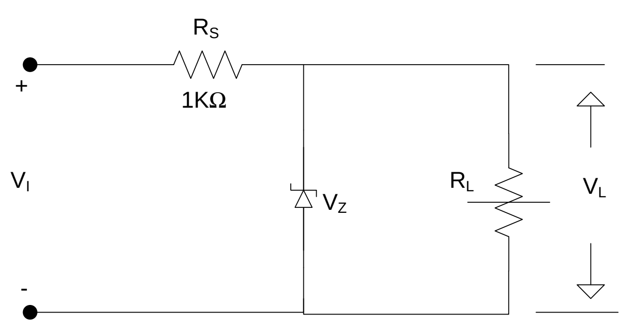

Fig .1.1:

The voltage regulator circuit of figure 1.1 is required to maintain the load voltage V_L at 6V. The circuit derives its power from a supply of V_I = 30V. The breakdown voltage of the zener diode,V_Z = 6V, and the maximum current that can be permitted to flow through the zener diode is =12mA.

4. Determine the minimum value of R_{Lmin} which will ensure this requirement.
(Take I_{Zmin} = 10% of I_{Zmax}).

(a) 500Ω
(b) 563Ω
(c) 263Ω
(d) 632Ω
(e) None of the above.

5. Determine the maximum value of R_{Lmax} which will ensure this requirement.
(a) 500Ω
(b) 563Ω
(c) 263Ω
(d) 632Ω
(e) None of the above.

6. Determine the minimum value of the power drop across the zener diode.
(a) 2.350W
(b) 0.3156W
(c) 3.156W
(d) 4.012W
(e) None of the above.

7. Determine the maximum value of the power drop across the zener diode.
(a) 2.350W
(b) 0.3156W
(c) 3.156W
(d) 4.012W
(e) None of the above.

8. Which of the following is not true?
(a) A transformer is usually employed in order to step-down the supply voltage
(b) A transformer is usually employed to protect from shocks.
(c) When the sinusoidal input voltage goes positive, the diode is forward-biased in a half wave rectifier.
(d) For an ideal diode, forward resistance r_f =100%.
(e) None of the above.

9. Find the average value of the full-wave rectified output voltage if the input voltage is 45V.
(a) 3.86478V
(b) 28.6478V
(c) 38.6478V
(d) 2.8478V
(e) None of the above.

10. Find the efficiency of a half wave rectifier with $\frac{r_f}{R_L} = 4$
(a) 1.0125%
(b) 8.9125%
(c) 8.10%
(d) 9.9125%
(e) None of the above.

11. Which of the following is not true?
(a) A transformer is usually employed in order to step-down the supply voltage
(b) A transformer is usually employed to protect from shocks.
(c) When the sinusoidal input voltage goes positive, the diode is forward-biased in a half wave rectifier.
(d) For an ideal diode, forward resistance r_f =100%.
(e) None of the above.

12. Find the average value of the full-wave rectified output voltage if the input voltage is 45V.
(a) 3.86478V
(b) 28.6478V
(c) 38.6478V
(d) 2.8478V

13. The movement of electrons and holes across a PN junction is known as
(a) Depletion Layer
(b) Potential Barrier
(c) Diffusion
(d) Electron and Holes equilibrium
(e) None of the above.

14. The external voltage required to overcome this barrier potential to allow electrons to move freely across the junction is dependent on the type of semiconductor material and temperature.
(a) true
(b) false
(c) unknown
(d) none of the above

15. The potential barrier on a PN Junction will always exist even if the device is not connected to any external power source.
(a) true
(b) false
(c) unknown
(d) none of the above

16. The external voltage applied to the PN Junction for biasing can:
(a) Eliminate the potential barrier.
(b) Increase the voltage of the potential barrier.
(c) Increases or Decrease the potential barrier.
(d) None of the above.

17. ………… is the forward voltage at which the diode starts conducting.
(a) Leakage voltage
(b) Breakdown voltage
(c) Knee voltage
(d) Biasing voltage
(e) None of the above

18. A turn-on voltage of a silicon diode is nearly
(a) 0.2 V
(b) 0.3 V
(c) 0.4 V
(d) 0.7 V
(e) None of the above

19. When forward bias is applied to a junction diode,
(a) it increases the potential barrier
(b) it decreases the potential barrier
(c) it reduces the majority-carrier current to zero
(d) it reduces the minority-carrier current to zero
(e) None of the above

20. When a PN-junction is biased in the forward direction
(a) only holes in the *P* region are injected into the *N* region
(b) only electrons in the *N* region are injected into the *P* region
(c) majority carriers in each region are injected into the other region
(d) no carriers move

21. Forward bias applied to a *PN*-junction diode is increased from zero to higher values. Rapid increase in the current flow for a relatively small increase in voltage occurs.
(a) immediately
(b) only after the forward bias exceeds the potential barrier
(c) when the flow of minority carriers is sufficient to cause an avalanche breakdown
(d) when the depletion area becomes larger than the space-charge area

22. In a full-wave rectifier, the current in each diode flows for
(a) the complete cycle of the input voltage
(b) only half-cycle of the input voltage
(c) less than half-cycle of the input voltage
(d) zero time

Using figure 1.2 below, answer question **23** to **27**

23. What is the block labeled A
(a) Filter
(b) Voltage Regulator
(c) Load
(d) Diode Rectifier
(e) None of the above

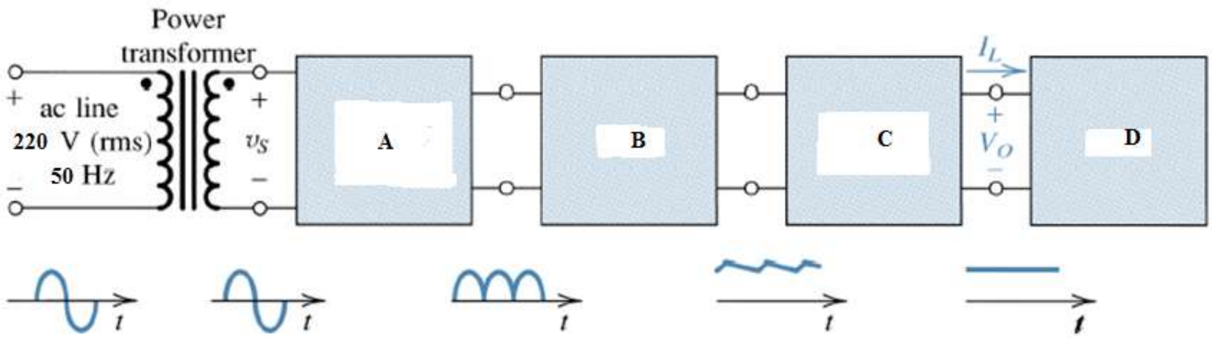

Figure 1.2

24. How would you describe the output of block A
(a) Midwave rectification
(b) Halfwave rectification
(c) Full wave rectification
(d) All of the above
(e) None of the above

25. What electronic component can be used to reduce the ripple in the output of block B
(a) Rippler
(b) Smoother
(c) Capacitor
(d) Inductor
(e) All of the above

26. Which of the following can be a typical block D device
(a) Iphone
(b) Samsung Galaxy S5
(c) Macbook Air
(d) All of the above
(e) None of the above

27. What electronic component can be used for a simple block C circuit
(a) Variable Resistor
(b) Zener Diode
(c) Capacitor
(d) IC
(e) None of the above

28. Which of the following is a mechanism by which mobile carriers move in a semiconductor?
(a) Drift
(b) Difference
(c) Dispersion
(d) Displacement
(e) Shift

29. Which of the following is not a biasing state of a PN junction diode?
(a) Forward
(b) Reserve
(c) Reverse
(d) Zero
(e) None of the above

30. A P-type semiconductor has
(a) Electrons majority Carriers and Holes minority carriers
(b) Holes majority Carriers and Electrons minority carriers
(c) Electrons and Holes majority Carriers
(d) Electrons and Holes minority carriers
(e) None of the above

31. When a PN junction diode is reversed biased all of the following happens except
(a) Depletion region widens
(b) Depletion region resistance becomes high
(c) Leakage current flows
(d) Minority Carriers cross the depletion region
(e) None of the above

32. In …………….the electron and hole concentrations are equal because carriers are created in pairs
(a) intrinsic semiconductors
(b) extrinsic semiconductors
(c) intrinsic conductors
(d) extrinsic conductors
(e) None of the above

33. ………..is the process of introducing an impurity material to a pure semiconductor material to have a p-type or an n-type material.
(a) Bonding
(b) Biasing
(c) Doping
(d) Conduction
(e) None of the above

34. Major part of electric current in an intrinsic semiconductor is due to drift of
(a) Conduction –band electrons
(b) valence –band electrons
(c) conduction-band holes
(d) valence-band holes

35. When the temperature of an intrinsic semiconductor is raised,
(a) Its resistance increases
(b) its conductance increases
(c) holes are created in conduction band
(d) Fermi energy level shifts upwards

36. The movement of a hole in a semiconductor is brought about by
(a) The movement of the atomic core
(b) the atomic core changing from +4 charge to +3 charge
(c) The vacancy being filled by a free-electrons
(d) the vacancy being filled by valence electron from a neighboring atom

The diodes used in a bridge rectifier circuit have the following parameters:

$V_T = 0V$, $r_f = 10\Omega$, $R_R = \infty$, V_Z = very large

The secondary winding of the transformer has a resistance $R_S = 10\Omega$. A load of 970Ω is connected across its output. If the voltage across the secondary is $v_s = 20 \sin wt$ volts, determine
Use the above information to solve question **37** to **40**

37. PIV rating of the diodes
(a) 10.0V
(b) 20.0V
(c) 12.7V
(d) 157.5V
(e) None of the above.

38. dc current through the load.
(a) 10.0mA
(b) 20.0mA
(c) 12.7mA
(d) 157.5mA
(e) None of the above.

39. dc power supplied to the load.
(a) 10.0mW
(b) 20.0mW
(c) 12.7mW
(d) 157.5mW
(e) None of the above.

40. Conversion efficiency of the current.
(a) 79.56%
(b) 60.0%
(c) 72.7%
(d) 85.75%
(e) None of the above.